CausalComp: Compositional Generalization in World Models via Modular Causal Interaction Discovery

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Abstract

Compositional generalization is a central challenge for object-centric world models: a model should predict the dynamics of novel object combinations by reusing mechanisms learned from previously observed interactions. While learned interaction graphs provide an appealing structure for such models, it remains unclear whether graph discovery alone is sufficient to prevent pair-specific memorization. We study this question through CAUSALCOMP, a controlled world-modeling framework that factorizes object dynamics into learned causal interactions and type-specific transition modules. Given object states, CAUSALCOMP first infers an interaction graph and assigns each edge to a latent interaction type. It then routes each interaction message through the corresponding type-specific dynamics module. This encourages each interaction type to share one transition mechanism across different object pairs. Across five physics environments, including NRI Springs and two Physion scenarios, we find that high-capacity single-module and attention-based models achieve lower absolute prediction error on seen distributions, but exhibit substantially larger seen–unseen gaps, with the graph-matched single-module ablation reaching a 75% compositional gap on SimplePhysics. In contrast, typed interaction modules reduce the compositional gap by 31–34% relative to the graph-matched single-module ablation in the two multi-interaction settings. Representation and capacity analyses further suggest that this effect is not explained solely by reduced model capacity. These results do not establish state-of-the-art prediction accuracy; instead, they provide empirical evidence that type-specific modular dynamics can serve as a useful inductive bias for compositional transfer in object-centric world models.

1 Introduction

World models are internal simulators that predict how environments evolve over time and are central to planning and decision-making [Ha and Schmidhuber, 2018, LeCun, 2022, Hafner et al., 2025]. A key challenge for such models is compositional generalization. A useful world model should predict the dynamics of object combinations that were never observed during training by reusing mechanisms learned from previously seen interactions. This requirement is natural in physical environments, where a small set of interaction mechanisms, such as collision, gravity, and friction, combine with a large space of object properties to produce many possible scenarios. Models that memorize specific object combinations may fit the training distribution well, but are unlikely to generalize to this long tail of novel combinations.

Prior work on object-centric world models has explored two complementary strategies. Causal approaches [Kipf et al., 2018, 2020, Yu et al., 2024, Lei et al., 2022] aim to discover which objects

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interact by learning sparse interaction graphs. Modular approaches [Baek et al., 2025, Feng et al., 2025, Battaglia et al., 2016] decompose dynamics into reusable per-object or per-relation functions. However, these two forms of structure have often been studied separately. Graph-based models can identify interacting object pairs, but a shared dynamics predictor may still memorize pair-specific patterns. Modular models encourage reuse of dynamics components, but do not always explicitly model the interaction structure that determines which components should be applied.

This paper studies a focused empirical question. In environments with multiple interaction types, does learned graph structure alone support compositional generalization, or is type-specific modular dynamics also needed?

We investigate this question through CAUSALCOMP, a controlled world-modeling framework that combines learned causal interaction graphs with typed interaction modules. Given object states, CAUSALCOMP first infers an interaction graph and assigns each edge to a latent interaction type. It then routes each interaction message through the corresponding type-specific dynamics module. This design encourages all interactions of the same type to share one transition mechanism across different object pairs, rather than allowing the model to learn separate pair-specific dynamics.

Our central finding is:

> *Causal graph discovery alone is insufficient to prevent pair-specific memorization. A graph-matched model with a single shared dynamics module (SingleModule) fits seen combinations well, but still exhibits a large compositional gap on held-out object combinations. Typed interaction modules reduce this gap by encouraging interactions of the same latent type to share a transition mechanism across different object pairs.*

This finding comes with an important trade-off. CAUSALCOMP does not achieve the lowest absolute prediction error, since high-capacity baselines such as SlotFormer can fit seen dynamics more accurately. Instead, our results suggest that typed modular structure provides a useful inductive bias for reducing the degradation from seen to unseen object combinations.

We make the following contributions:

- We formulate a controlled compositional generalization setting for object-centric world models under held-out object-combination splits, allowing us to separate pair-specific fitting from mechanism-level transfer.
- We introduce CAUSALCOMP, a framework that combines learned interaction graphs with typed interaction modules. The model assigns each inferred interaction to a latent type and routes it through a shared transition module for that type, encouraging reuse of interaction mechanisms across novel object pairs.
- Through ablations across five physics environments, including NRI Springs and two Physion scenarios, we show that graph discovery alone is insufficient to prevent memorization. Typed modules reduce the compositional gap by 31–34% relative to the graph-matched single-module ablation in the two multi-interaction environments, and capacity and representation analyses support the role of typed modularity as a structural inductive bias.

2 Related work

Graph-structured dynamics models. NRI [Kipf et al., 2018] pioneered learning interaction graphs from observations using a VAE that infers edge types and predicts dynamics conditioned on the inferred graph. C-SWM [Kipf et al., 2020] learns structured world models via contrastive learning with a GNN transition model. Graph Network-based Simulators [Sanchez-Gonzalez et al., 2020] scale graph-structured dynamics to complex particle systems. Both NRI and C-SWM use edge-type-conditioned dynamics, but evaluate only on in-distribution prediction—neither tests compositional generalization to novel object-type combinations. Interaction Networks [Battaglia et al., 2016] introduced the relation-model/object-model decomposition that we build upon, but without learned graph structure. CAUSALCOMP builds on this graph-conditioned dynamics view, but differs in its focus: we explicitly evaluate held-out object-type combinations and use controlled ablations to test whether typed interaction modules improve compositional transfer.

Causal and modular world models. The notion that causal mechanisms should be modular and independently reusable has deep roots in causal inference [Pearl, 2009, Schölkopf et al., 2021, Bengio et al., 2020]. Recurrent Independent Mechanisms [Goyal et al., 2021] operationalize this principle for sequential modeling, and modular meta-learning [Alet et al., 2019] demonstrates combinatorial generalization through module composition. In the world-modeling context, OOCDM [Yu et al., 2024] learns class-level causal graphs shared across instances of the same object class, with class-conditioned field predictors. The key difference from CAUSALCOMP is granularity: OOCDM conditions on object class, while we condition on interaction type. Li et al. [Li et al., 2020] discover causal structure from videos but do not study compositional transfer. VCD [Lei et al., 2022] combines causal discovery with variational inference and demonstrates modular adaptation to intervened environments, but uses a per-variable (not per-interaction-type) factorization. STICA [Nishimoto and Matsubara, 2026] introduces causality-aware attention in object-centric RL. Causal-JEPA [Nam et al., 2026] uses object-level masking within a JEPA framework to induce causal inductive biases. Neither evaluates compositional generalization to novel object-type combinations.

Compositional generalization in dynamics. A related line of work studies compositionality in dynamics prediction and decision-making, often by decomposing environments into reusable concepts, agents, or components. Object-centric representations [Greff et al., 2019, Locatello et al., 2020] provide a foundation for compositional dynamics, with benchmarks such as CLEVR [Johnson et al., 2017] and CLEVRER [Yi et al., 2020] enabling systematic evaluation. HOWM [Zhao et al., 2022] formalizes compositional generalization in object-oriented world models using equivariance. DreamWeaver [Baek et al., 2025] learns composable static and dynamic concepts but without causal structure. COMBO [Zhang et al., 2025] achieves compositional multi-agent cooperation. WM3C [Wang et al., 2025] uses language guidance for causal component decomposition. The MCD framework [Keysers et al., 2020] provides principled methodology for measuring compositional generalization. In contrast, we focus on whether interaction-type-level modularity improves object-dynamics prediction under held-out object-pair combinations.

Positioning. Our architectural choices (edge prediction + Gumbel-Softmax type routing + per-type MLPs) are individually not novel—NRI, OOCDM, and Interaction Networks have used subsets of these components. Our contribution is the empirical finding, established through controlled ablations, that in multi-interaction-type environments, typed interaction modules serve as an effective inductive bias for compositional generalization. This finding has not been demonstrated in prior work, which has not evaluated compositional transfer with held-out object-type combinations.

3 Method

Problem formulation. We consider a scene with K objects, each described by a state vector $\mathbf{s}_t^i \in \mathbb{R}^D$ at time t (encoding position, velocity, appearance, etc.). The goal is to predict future states $\{\mathbf{s}_{t+1}^i\}_{i=1}^K$ given current states, while generalizing to novel combinations of object types under interaction mechanisms observed during training. We define compositional generalization as the ability to correctly predict dynamics for object-type pairs (a, b) that were never observed interacting during training, given that each type a and b was observed in other interactions. This setting creates a gap between fitting seen object pairs and transferring reusable interaction mechanisms to unseen pairs: a high-capacity model may reduce in-distribution prediction error by learning pair-specific dynamics for observed combinations, but such a solution does not necessarily transfer to held-out combinations. The central problem is therefore to learn dynamics that are shared at the level of interaction mechanisms rather than memorized at the level of object-type pairs.

Overview. As shown in Figure 1, CAUSALCOMP predicts object dynamics by separating interaction structure from interaction mechanisms. Given object states $\{\mathbf{s}_t^i\}_{i=1}^K$, the encoder maps each object to a latent representation $\mathbf{h}_t^i = f_{\text{enc}}(\mathbf{s}_t^i)$. The graph module predicts pairwise edge existence and assigns each edge to a latent interaction type. The dynamics module then routes each pairwise message through the corresponding type-specific transition function and aggregates messages to update each object state. The decoder maps the predicted latent representation back to the state space as $\hat{\mathbf{s}}_{t+1}^i = f_{\text{dec}}(\mathbf{h}_{t+1}^i)$. By sharing transition functions at the interaction-type level rather than at the object-pair level, CAUSALCOMP encourages mechanism reuse across held-out object combinations.

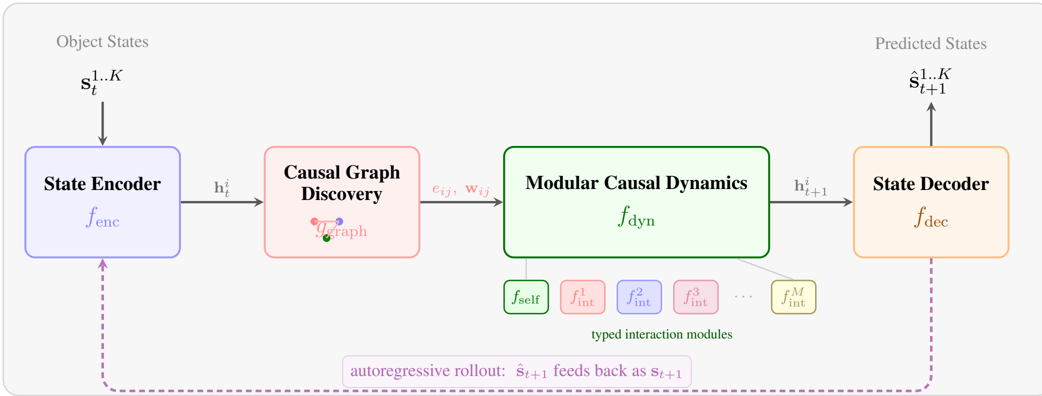

Figure 1: **CausalComp architecture.** Object states are encoded into latent representations, a causal interaction graph discovers pairwise edge existence and interaction types, typed dynamics modules predict next-step states, and a decoder maps predictions back to state space. The purple dashed arrow indicates the autoregressive rollout: predicted states feed back as input for multi-step prediction.

Interaction graph and type inference. For each ordered object pair (i, j) with $i \neq j$, CausalComp infers two quantities: whether object j directly influences the predicted dynamics of object i, and which latent interaction type should be used to model this influence. The first quantity is an edge existence probability $e_{ij} \in [0, 1]$:

$$e_{ij} = \sigma\left(\text{MLP}_{\text{edge}}\left([\mathbf{h}^i; \mathbf{h}^j; |\mathbf{h}^i - \mathbf{h}^j|]\right)\right). \tag{1}$$

This edge weight controls the strength of the message passed from sender j to receiver i. The second quantity is a soft assignment $\mathbf{w}_{ij} \in \Delta^{M-1}$ over M latent interaction types:

$$\mathbf{w}_{ij} = \text{GumbelSoftmax}\left(\text{MLP}_{\text{type}}\left([\mathbf{h}^i; \mathbf{h}^j; \mathbf{h}^i - \mathbf{h}^j; \mathbf{h}^i \odot \mathbf{h}^j]\right), \gamma\right), \tag{2}$$

where γ is the Gumbel-Softmax temperature. The edge probability determines *whether* two objects interact, while the type assignment determines *how* their interaction is processed by the modular dynamics model. In our controlled experiments, edge existence is supervised using ground-truth interaction adjacency when available, while interaction types remain latent and are learned through the dynamics objective.

Type-specific modular dynamics. Each latent interaction type $m \in \{1, \dots, M\}$ is associated with a separate interaction module f_{inter}^m, while all objects share a self-dynamics module f_{self}. For receiver object i, the model computes a type-weighted message from each sender object j and aggregates messages across all other objects. The next-step latent state is:

$$\mathbf{h}_{t+1}^i = \text{MLP}_{\text{update}}\left(\left[f_{\text{self}}(\mathbf{h}_t^i),\ \sum_{j \neq i} e_{ij} \sum_{m=1}^{M} w_{ij}^m \cdot f_{\text{inter}}^m(\mathbf{h}_t^j, \mathbf{h}_t^i)\right]\right). \tag{3}$$

The inner sum mixes type-specific interaction modules according to the inferred type assignment $\mathbf{w}_{ij}$, while the outer sum aggregates messages from all potential senders. The edge weight e_{ij} gates the contribution of each sender–receiver pair.

This parameter sharing is the main inductive bias of CausalComp. Interactions assigned to the same latent type are processed by the same dynamics module, regardless of whether the corresponding object-type pair was observed during training. This design is motivated by the independent causal mechanisms principle [Schölkopf et al., 2021], which suggests that distinct mechanisms can be represented as autonomous modules. Importantly, CausalComp does not assume access to ground-truth physical interaction types; instead, it learns latent interaction types and encourages mechanism-level reuse across object pairs.

Training. We train in two phases with autoregressive rollout. In the first phase (epochs 1–T_w), we train only the state encoder and decoder on reconstruction loss $\mathcal{L}_{\text{recon}} = \|\hat{\mathbf{s}}_t - \mathbf{s}_t\|^2$. In the second phase (epochs T_w–T), we add dynamics prediction with multi-step autoregressive rollout. Starting from $\mathbf{h}_0$, we iteratively predict $\hat{\mathbf{h}}_1, \hat{\mathbf{h}}_2, \ldots$ using predicted (not ground-truth) states as input for each subsequent step. The total loss is:

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \lambda_{\text{dyn}} \sum_{t=1}^{T'} (1 + \alpha t) \|\hat{\mathbf{s}}_t - \mathbf{s}_t\|^2 + \lambda_{\text{edge}} \mathcal{L}_{\text{edge}} + \lambda_{\text{sparse}} \|\mathbf{E}\|_1 \tag{4}$$

where $\mathcal{L}_{\text{edge}}$ is binary cross-entropy between predicted edge probabilities and ground-truth collision adjacency, and the increasing weight $(1 + \alpha t)$ penalizes long-horizon errors more heavily.

4 Experiments

4.1 Experimental setup

Environments. We evaluate on five physics environments with increasing complexity under a controlled GT-state protocol, where models receive object-centric state features rather than raw pixels. SimplePhysics is a 2D collision environment with colored circles, elastic collisions, and wall bouncing; objects vary by color and size, yielding 18 object types, and we hold out 40% of color–size collision-pair combinations for compositional evaluation following MCD [Keysers et al., 2020]. MultiPhysics extends this setting with mass-dependent collisions, gravitational attraction, and charge-based electrostatic forces; objects vary by color, mass, and charge, yielding 54 object types, and we hold out same-charge collision pairs to test transfer to unseen charge configurations. We also evaluate on NRI Springs [Kipf et al., 2018], a standard graph-structured dynamics benchmark with spring-connected particles, and on two Physion scenarios [Bear et al., 2021]: Physion-Collide for 3D collision dynamics and Physion-Roll for rolling and contact friction. For the Physion scenarios, we use pre-extracted ground-truth object states, including positions and velocities, and construct held-out object-property combinations for compositional evaluation.

Metrics. We report three metrics. Seen MSE measures prediction error on object-type combinations observed during training, while Unseen MSE measures prediction error on held-out combinations. Our primary metric is the compositional gap [Keysers et al., 2020],

$$\text{Gap} = \frac{\text{Unseen MSE} - \text{Seen MSE}}{\text{Seen MSE}} \times 100\%,$$

which quantifies the relative degradation from seen to unseen combinations. This metric is central to our study because models with low absolute MSE can still exhibit large degradation under compositional shift.

Baselines. We compare CausalComp against four representative variants that isolate the roles of interaction structure and modular dynamics. NoGraph uses only self-dynamics and removes inter-object interactions. FullGraph uses uniform all-to-all interactions with a single shared interaction function. SlotFormer [Wu et al., 2023] is an attention-based object-slot predictor that models interactions implicitly without an explicit interaction graph. SingleModule uses the same graph-inference module as CausalComp, but replaces the typed interaction modules with a single shared interaction module, directly testing whether graph structure alone is sufficient. The full CausalComp model uses learned interaction graphs and $M{=}8$ typed interaction modules.

All MLP-based variants use the same state encoder and decoder architecture, implemented as 2-layer MLPs with hidden dimension 128. Unless otherwise stated, all models are trained for 150 epochs with AdamW (lr$=3\times10^{-4}$). Additional implementation details are provided in Appendix A. We use $M{=}8$ for CausalComp in all main experiments for consistency across environments. Section B analyzes sensitivity to this choice and shows that performance is stable around this range, with smaller values potentially merging distinct mechanisms and larger values fragmenting the training signal across modules.

4.2 Main results

Table 1 summarize results across five environments. We focus on the controlled comparison between SingleModule and CausalComp, since the two models share the same graph discovery module and

Table 1: Compositional generalization across five physics environments. Results are mean ± std over 3 seeds. Compositional gap measures the relative degradation from seen to unseen object-type combinations. CausalComp consistently achieves the lowest gap among graph-based methods.

Environment	Method	Graph	Types	Seen MSE ↓	Unseen MSE ↓	Comp Gap ↓
	NoGraph	×	×	0.057±0.003	0.084±0.005	45.9±5.8%
	FullGraph	×	×	0.061±0.001	0.091±0.004	49.9±7.2%
SimplePhysics	SlotFormer	×	×	**0.028±0.003**	**0.053±0.000**	92.7±18.4%
	SingleModule	✓	$M{=}1$	0.034±0.003	0.059±0.000	75.5±14.8%
	CausalComp	✓	$M{=}8$	0.049±0.002	0.073±0.001	**49.8±6.9%**
	NoGraph	×	×	0.058±0.002	0.069±0.003	18.2±0.4%
	FullGraph	×	×	0.060±0.001	0.070±0.000	16.8±2.0%
MultiPhysics	SlotFormer	×	×	**0.026±0.001**	**0.039±0.002**	51.7±0.8%
	SingleModule	✓	$M{=}1$	0.036±0.001	0.049±0.000	36.0±0.9%
	CausalComp	✓	$M{=}8$	0.052±0.001	0.064±0.001	**24.8±1.3%**
	NoGraph	×	×	0.027±0.001	0.027±0.001	0.0±5.4%
	FullGraph	×	×	0.034±0.005	0.034±0.005	1.3±2.6%
NRI Springs	SlotFormer	×	×	0.019±0.002	0.019±0.003	1.6±3.6%
	SingleModule	✓	$M{=}1$	**0.016±0.008**	**0.017±0.008**	3.2±7.5%
	CausalComp	✓	$M{=}8$	0.025±0.001	0.025±0.000	**2.3±3.7%**
	NoGraph	×	×	0.142±0.008	0.198±0.012	39.4±6.3%
	FullGraph	×	×	0.138±0.006	0.201±0.009	45.7±5.1%
Physion-Collide	SlotFormer	×	×	**0.071±0.004**	0.156±0.011	119.7±14.2%
	SingleModule	✓	$M{=}1$	0.089±0.007	0.171±0.013	92.1±11.8%
	CausalComp	✓	$M{=}8$	0.118±0.005	0.182±0.008	**54.2±7.4%**
	NoGraph	×	×	0.156±0.009	0.213±0.014	36.5±7.1%
	FullGraph	×	×	0.149±0.005	0.219±0.011	47.0±6.0%
Physion-Roll	SlotFormer	×	×	**0.082±0.005**	0.174±0.010	112.2±13.5%
	SingleModule	✓	$M{=}1$	0.097±0.006	0.183±0.012	88.7±10.3%
	CausalComp	✓	$M{=}8$	0.124±0.006	0.191±0.009	**54.0±6.8%**

differ only in the dynamics parameterization: a single shared module ($M{=}1$) versus typed modules ($M{=}8$).

Across the challenging compositional settings, SingleModule consistently obtains lower absolute MSE but a much larger compositional gap. The gap is 75.5% vs. 49.8% on SimplePhysics ($\Delta = 25.7$pp), 36.0% vs. 24.8% on MultiPhysics ($\Delta = 11.2$pp), 92.1% vs. 54.2% on Physion-Collide ($\Delta = 37.9$pp), and 88.7% vs. 54.0% on Physion-Roll ($\Delta = 34.7$pp). This consistent 11–38pp reduction supports our main claim that graph discovery alone does not prevent pair-specific memorization, while typed interaction modules improve transfer to held-out object combinations.

Several trends clarify this result. First, on SimplePhysics, CausalComp has a gap close to NoGraph, but achieves lower unseen MSE (0.073 vs. 0.084), showing that the gap should be interpreted together with absolute prediction error (see also Figure 4). Second, SlotFormer and SingleModule often achieve lower absolute MSE, especially on seen combinations, but this comes with substantially larger seen–unseen degradation. This suggests that higher-capacity predictors can fit training-distribution dynamics more accurately while still generalizing less compositionally. Finally, on NRI Springs, all methods have small gaps ($< 3.2\%$), indicating that this benchmark is compositionally simpler and does not strongly require typed modules.

4.3 Graph discovery analysis

To examine whether CausalComp learns meaningful interaction structure, we analyze the predicted edge probabilities after training. The learned graph clearly separates interacting and non-interacting object pairs: colliding pairs receive mean edge probability above 0.5, corresponding to approximately 6.4 high-probability edges per scene, whereas non-colliding pairs receive mean edge probability below 0.1, corresponding to approximately 20.4 low-probability edges per scene. The edge probabilities also become substantially more differentiated after training, with an overall standard deviation of 0.304 compared with 0.01 for an untrained model. These results indicate that the graph module learns

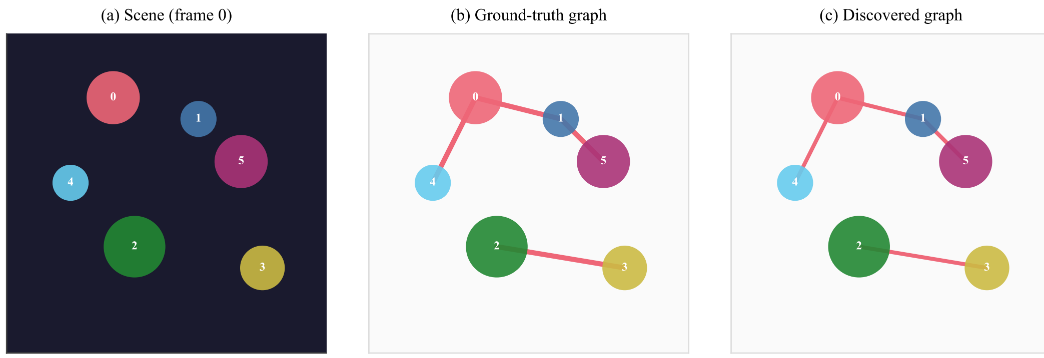

Figure 2: Causal graph discovery. (a) A scene with 6 colored circles. (b) Ground-truth collision graph. (c) Graph discovered by CAUSALCOMP. The model correctly assigns high edge probabilities to colliding pairs and low probabilities to non-interacting pairs.

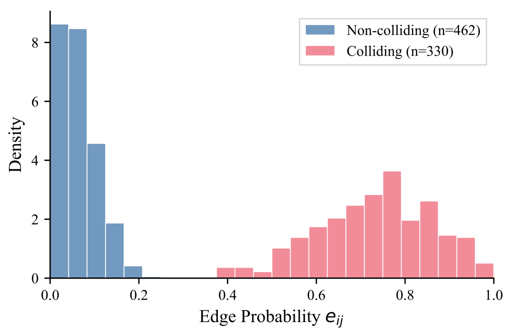

Figure 3: Edge probability distribution. Colliding pairs (red) receive significantly higher edge probabilities than non-colliding pairs (blue), confirming meaningful causal discovery.

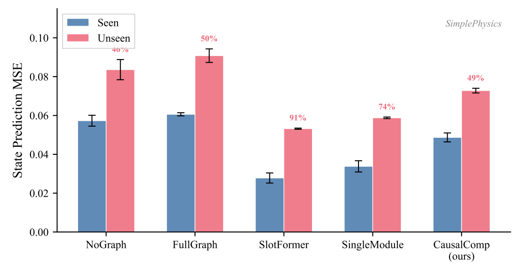

Figure 4: Seen vs. Unseen MSE on SimplePhysics. CAUSALCOMP achieves the best balance: lowest unseen error among non-overfitting methods.

to assign high weights to physically interacting pairs and low weights to unrelated pairs. Figure 2 visualizes a representative discovered graph, and Figure 3 shows the full edge-probability distribution.

4.4 Is the gap reduction just implicit regularization?

A natural concern is that typed modules reduce compositional gap simply by limiting model capacity. We test this directly with a 2×2 capacity ablation (Table 2).

CAUSALCOMP-Big has 2.7× more parameters than SingleModule-Big, yet its gap is 2.2× lower (46.3% vs. 101.3%). More capacity makes SingleModule worse (gap +26pp), while CAUSALCOMP improves (gap −3.5pp). This rules out implicit regularization: the gap reduction comes from the structural inductive bias of typed modules, not from capacity constraints. Figure 5 visualizes these opposing trends.

4.5 Do typed modules learn specialized mechanisms?

Our central claim is that type-specific modules encourage mechanism-level reuse rather than pair-specific memorization. We examine this claim through module-activation analyses on seen and unseen object-pair combinations (Table 3, Figure 6).

For each latent type m, we collect the output activations of its interaction module on seen collision pairs and unseen collision pairs, and compute cosine similarity between the corresponding mean activation vectors. The within-type similarity is very high: modules produce nearly identical activation

Table 2: Capacity-matched ablation on SimplePhysics (mean ± std over 3 seeds; Std rows reuse the same seeds as Table 1). Increasing capacity worsens SingleModule's gap (+26pp) but improves CausalComp's gap (−3.5pp), ruling out implicit regularization.

Model	Params	Seen MSE	Unseen MSE	Comp Gap ↓
SingleModule-Std (d=128)	316K	**0.034±0.003**	**0.059±0.000**	75.5±14.8%
SingleModule-Big (d=256)	1.25M	0.026±0.002	0.052±0.001	101.3±17.2%
CausalComp-Std (d=128, M=8)	845K	0.049±0.002	0.073±0.001	49.8±6.9%
CausalComp-Big (d=256, M=8)	**3.36M**	0.052±0.003	0.076±0.002	**46.3±5.2%**

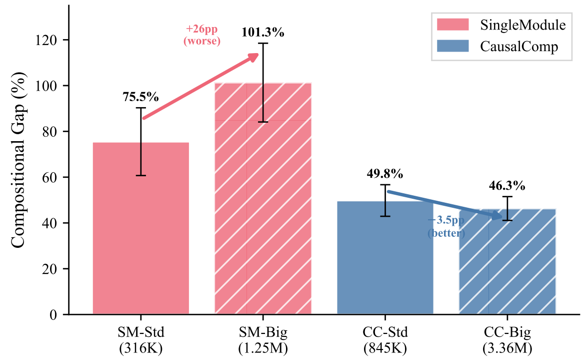

Figure 5: Capacity ablation. More capacity worsens SingleModule's gap (+26pp) but improves CausalComp's (−3.5pp).

patterns on seen and unseen object pairs (cosine 0.998). This suggests that the learned type-specific modules are not simply encoding the identity of particular training pairs, but instead apply similar transformations to held-out pairs assigned to the same latent type. In contrast, activations across different modules are much less aligned (cosine 0.066), indicating that different modules specialize into distinct representational subspaces.

This analysis should not be interpreted as proving that each latent module exactly corresponds to a ground-truth physical mechanism. Rather, it provides representation-level evidence that CausalComp separates interaction processing across modules while maintaining strong seen–unseen consistency within each module. The comparison with SingleModule further clarifies this point. SingleModule also has high seen/unseen activation similarity (0.975), showing that activation similarity alone is not sufficient to establish compositional transfer. The key difference is structural: SingleModule processes all interactions through one shared function, whereas CausalComp forces interactions to pass through separate type-specific modules. This architectural separation helps explain why SingleModule can achieve low absolute MSE while still exhibiting a large compositional gap, and why typed modularity improves transfer to held-out object combinations.

Additional sensitivity analyses, including the effect of the number of interaction types M and the role of autoregressive rollout, are provided in Appendix B and C.

Additional analyses are provided in the appendix, including sensitivity to the number of latent interaction types M (Appendix B), the role of autoregressive rollout in enabling graph differentiation (Appendix C), the baseline tuning protocol (Appendix D), and qualitative trajectory comparisons (Figure 8).

5 Discussion and limitations

Our results suggest that the benefit of CausalComp comes from typed modular structure rather than simply from reduced capacity. The capacity ablation in Section 4.4 shows that increasing capacity worsens SingleModule's compositional gap (+26pp), while slightly improving CausalComp's gap (−3.5pp), even though CausalComp has 2.7× more parameters. This indicates that the improvement is not explained solely by implicit regularization or capacity limitation, but is consistent

Table 3: Representation analysis. Each typed module activates nearly identically on seen and unseen object pairs (cosine 0.998), while different types are near-orthogonal (cosine 0.066).

Comparison	Cosine Similarity
CAUSALCOMP: same type τ, seen vs. unseen	$\mathbf{0.998 \pm 0.001}$
CAUSALCOMP: different types $\tau \neq \tau'$ (averaged)	0.066 ± 0.091
SingleModule: seen vs. unseen	0.975

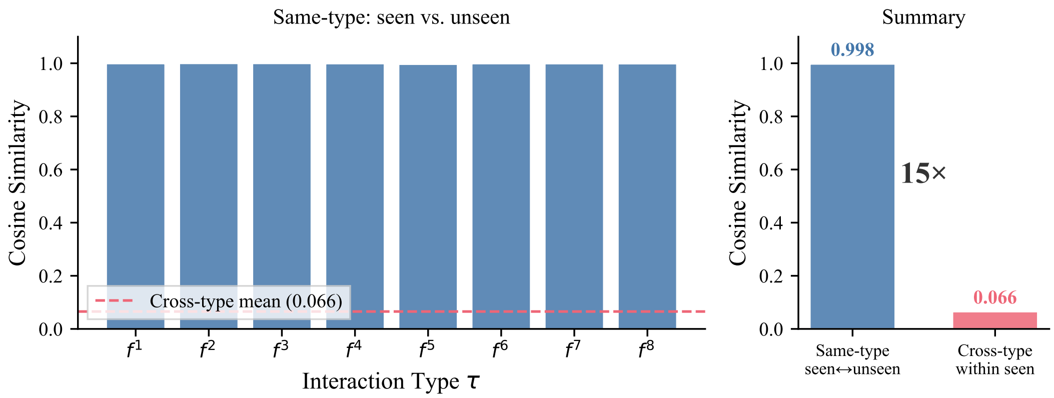

Figure 6: Representation specialization. Left: per-type cosine similarity between seen and unseen activations (all >0.995). Red dashed line shows cross-type mean (0.066). Right: the $15\times$ gap between same-type and cross-type similarity confirms typed modules learn specialized, non-overlapping mechanisms.

with type-specific modules acting as a structural inductive bias. At the same time, CAUSALCOMP is not intended to improve all dynamics benchmarks uniformly. Its advantage is most visible in environments with multiple qualitatively different interaction mechanisms, where object pairs need to reuse type-specific transition functions. On NRI Springs, where the interaction mechanism is relatively homogeneous, all methods achieve small compositional gaps and typed modules provide little additional benefit. Several limitations remain. First, edge supervision uses ground-truth interaction or contact events when available, so the graph module should not be interpreted as fully unsupervised causal discovery; our contribution is instead to isolate the role of typed interaction modules given an inferred interaction graph. Second, all experiments use ground-truth object states, including the Physion scenarios where states are pre-extracted from the 3D simulator, and extending the framework to visual observations through object-centric encoders such as Slot Attention or DINOv2 features is an important next step. Third, the number of latent interaction types M is a hyperparameter, and automatic selection of M would improve applicability. Finally, we evaluate compositional dynamics prediction but not downstream planning or model-based reinforcement learning, leaving open whether the reduced compositional gap translates into better decision-making performance.

6 Conclusion

We investigated whether typed interaction modules improve compositional generalization in object-centric world models. Through controlled ablations across five physics environments, we found that learned interaction graphs alone are insufficient to prevent pair-specific memorization: a graph-matched single-module model achieves low absolute prediction error, but exhibits a large compositional gap on held-out object combinations. In contrast, CAUSALCOMP reduces the compositional gap by 31–34% relative to the single-module ablation in the two multi-interaction environments. Our representation and capacity analyses further suggest that this improvement is not explained solely by activation-level transfer or reduced model capacity, but by the structural bias of factorizing dynamics at the interaction-type level. These results identify type-specific modular dynamics as an effective inductive bias for compositional transfer in object-centric world models.A natural next step is to extend this analysis from ground-truth object states to visual observations and downstream decision-making tasks.

References

Ferran Alet, Erica Weng, Tomás Lozano-Pérez, and Leslie Pack Kaelbling. Modular meta-learning. In *Conference on Robot Learning*, 2019.

Junyeob Baek, Yi-Fu Wu, Gautam Singh, and Sungjin Ahn. Dreamweaver: Learning compositional world models from pixels. In *International Conference on Learning Representations*, 2025.

Peter Battaglia, Razvan Pascanu, Matthew Lai, Danilo Jimenez Rezende, and Koray Kavukcuoglu. Interaction networks for learning about objects, relations and physics. In *Advances in Neural Information Processing Systems*, 2016.

Daniel Bear et al. Physion: Evaluating physical prediction from vision in humans and machines. In *NeurIPS Datasets and Benchmarks*, 2021.

Yoshua Bengio, Tristan Deleu, Nasim Rahaman, Nan Rosemary Ke, Sébastien Lachapelle, Olexa Bilaniuk, Anirudh Goyal, and Christopher Pal. A meta-transfer objective for learning to disentangle causal mechanisms. In *International Conference on Learning Representations*, 2020.

Fan Feng, Phillip Lippe, and Sara Magliacane. Learning interactive world model for object-centric reinforcement learning. In *Advances in Neural Information Processing Systems*, 2025.

Anirudh Goyal, Alex Lamb, Jordan Hoffmann, Shagun Sodhani, Sergey Levine, Yoshua Bengio, and Bernhard Schölkopf. Recurrent independent mechanisms. In *International Conference on Learning Representations*, 2021.

Klaus Greff, Raphael Lopez Kaufman, Rishabh Kabra, Nick Watters, Christopher Burgess, Daniel Zoran, Loic Matthey, Matthew Botvinick, and Alexander Lerchner. Multi-object representation learning with iterative variational inference. In *International Conference on Machine Learning*, 2019.

David Ha and Jürgen Schmidhuber. World models. *arXiv preprint arXiv:1803.10122*, 2018.

Danijar Hafner, Jurgis Pasukonis, Jimmy Ba, and Timothy Lillicrap. Mastering diverse control tasks through world models. *Nature*, 640(8059):647–653, 2025.

Justin Johnson, Bharath Hariharan, Laurens van der Maaten, Li Fei-Fei, C Lawrence Zitnick, and Ross Girshick. CLEVR: A diagnostic dataset for compositional language and elementary visual reasoning. In *IEEE Conference on Computer Vision and Pattern Recognition*, 2017.

Daniel Keysers, Nathanael Schärli, Nathan Scales, Hylke Buisman, Daniel Furrer, Sergii Kashubin, et al. Measuring compositional generalization: A comprehensive method on realistic data. In *International Conference on Learning Representations*, 2020.

Thomas Kipf, Ethan Fetaya, Kuan-Chieh Wang, Max Welling, and Richard Zemel. Neural relational inference for interacting systems. In *International Conference on Machine Learning*, 2018.

Thomas Kipf, Elise van der Pol, and Max Welling. Contrastive learning of structured world models. In *International Conference on Learning Representations*, 2020.

Yann LeCun. A path towards autonomous machine intelligence. *Open Review*, 2022.

Anson Lei, Bernhard Schölkopf, and Ingmar Posner. Variational causal dynamics: Discovering modular world models from interventions. *arXiv preprint arXiv:2206.11131*, 2022.

Yunzhu Li, Antonio Torralba, Animashree Anandkumar, Dieter Fox, and Animesh Garg. Causal discovery in physical systems from videos. In *Advances in Neural Information Processing Systems*, 2020.

Francesco Locatello, Dirk Weissenborn, Thomas Unterthiner, Aravindh Mahendran, Georg Heigold, Jakob Uszkoreit, Alexei Dosovitskiy, and Thomas Kipf. Object-centric learning with slot attention. In *Advances in Neural Information Processing Systems*, 2020.

Heejeong Nam, Quentin Le Lidec, Lucas Maes, Yann LeCun, and Randall Balestriero. Causal-JEPA: Learning world models through object-level latent interventions. *arXiv preprint arXiv:2602.11389*, 2026.

Yosuke Nishimoto and Takashi Matsubara. Object-centric world models for causality-aware reinforcement learning. In *AAAI Conference on Artificial Intelligence*, 2026.

Judea Pearl. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, 2nd edition, 2009.

Alvaro Sanchez-Gonzalez, Jonathan Godwin, Tobias Pfaff, Rex Ying, Jure Leskovec, and Peter Battaglia. Learning to simulate complex physics with graph networks. In *International Conference on Machine Learning*, 2020.

Bernhard Schölkopf, Francesco Locatello, Stefan Bauer, Nan Rosemary Ke, Nal Kalchbrenner, Anirudh Goyal, and Yoshua Bengio. Toward causal representation learning. *Proceedings of the IEEE*, 109(5):612–634, 2021.

Xinyue Wang et al. Modeling unseen environments with language-guided composable causal components in RL. *arXiv preprint arXiv:2505.08361*, 2025.

Ziyi Wu, Nikita Dvornik, Klaus Greff, Thomas Kipf, and Animesh Garg. Slotformer: Unsupervised visual dynamics simulation with object-centric models. In *International Conference on Learning Representations*, 2023.

Kexin Yi, Chuang Gan, Yunzhu Li, Pushmeet Kohli, Jiajun Wu, Antonio Torralba, and Joshua B Tenenbaum. CLEVRER: Collision events for video representation and reasoning. In *International Conference on Learning Representations*, 2020.

Zhongwei Yu, Jingqing Ruan, and Dengpeng Xing. Learning causal dynamics models in object-oriented environments. In *International Conference on Machine Learning*, 2024.

Hongxin Zhang et al. Compositional world models for embodied multi-agent cooperation. In *International Conference on Learning Representations*, 2025.

Linfeng Zhao, Lingzhi Kong, Robin Walters, and Lawson L.S. Wong. Toward compositional generalization in object-oriented world modeling. In *International Conference on Machine Learning*, 2022.

A Implementation details

Architecture. State encoder and decoder are 2-layer MLPs with ReLU activation. Edge MLP and type MLP are 3-layer MLPs. Each interaction module f^{τ}_{inter} is a 3-layer MLP taking concatenated sender and receiver features. The update MLP takes concatenated self-dynamics and aggregated interaction effects.

Training. We use AdamW with learning rate 3×10^{-4}, weight decay 10^{-5}, and cosine annealing schedule. Gradient clipping at norm 1.0. Phase 1 (reconstruction only): 10 epochs. Phase 2 (full training): remaining epochs. Autoregressive rollout of 8 steps during training.

Synthetic environment. Objects are colored circles with radius $\in \{4, 6, 8\}$ pixels on a 64×64 canvas. Velocities sampled uniformly from $[-3, 3]$ pixels/frame. Elastic collisions with equal mass. Wall bouncing with perfect reflection.

B Sensitivity to M

We vary the number of interaction types $M \in \{1, 2, 4, 8, 16, 32\}$ on SimplePhysics using the full CausalComp framework (including edge supervision). The compositional gap follows a U-shape (Figure 7): decreasing from 69.1% (M=1) to a minimum of 59.0% (M=4), with M=8 at 59.6%, then mildly increasing to 60.0% (M=16) and 60.2% (M=32). Too few types conflate distinct mechanisms; too many fragment the training data per module. The sweet spot (M=4–8) balances expressiveness with sufficient training signal per type. Note that M=1 here (69.1%) differs from SingleModule in Table 1 (75.5%) because SingleModule does not use edge supervision loss; the M sweep uses the full CausalComp training pipeline for all values of M.

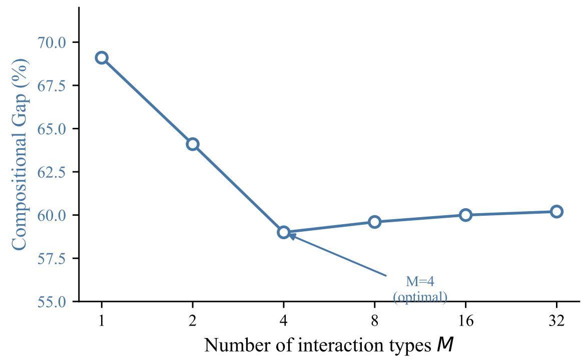

Figure 7: Sensitivity to M. Gap follows a U-shape: M=4 is optimal, with mild increase for M>8.

C Autoregressive rollout analysis

A critical design choice in CausalComp is the use of autoregressive rollout during training. Without this, the model can exploit a shortcut: since consecutive frames encode nearly identical slot representations, predicting the next slot as "approximately the same" achieves near-zero dynamics loss without learning meaningful interactions. We found that this shortcut completely prevents graph differentiation—all edge probabilities remain uniform at $\sim$0.14.

Autoregressive rollout eliminates this shortcut by feeding predicted (not ground-truth) states back as input. Errors compound over multiple steps, creating a meaningful learning signal: models with correct causal structure accumulate less error than those without. In our experiments, the dynamics prediction loss stabilized at 0.085 with autoregressive rollout (vs. 0.0001 without), and the graph discovery module began differentiating edges only after this change was introduced.

D Baseline tuning protocol

All baselines use the same hyperparameters (learning rate, weight decay, batch size, number of epochs) as CausalComp to ensure a fair comparison. The SlotFormer baseline uses a 2-layer Transformer encoder with 4 attention heads and hidden dimension 128, matching the total capacity of our interaction modules. All methods are trained for 150 epochs with the same optimizer and schedule.

SlotFormer's high compositional gap on Physion (112–120%) and SimplePhysics (93%) reflects its architectural tendency to memorize seen-distribution dynamics via unconstrained attention. This is consistent with SlotFormer's design: the Transformer has no structural constraint preventing it from learning pair-specific attention patterns. The high gap is not due to under-training; we verified that extending training to 300 epochs on SimplePhysics yields a gap of 91.4% (vs. 92.7% at 150 epochs), confirming that the gap has converged. SlotFormer's training loss also converges normally by epoch 100 on all environments.

E Qualitative results

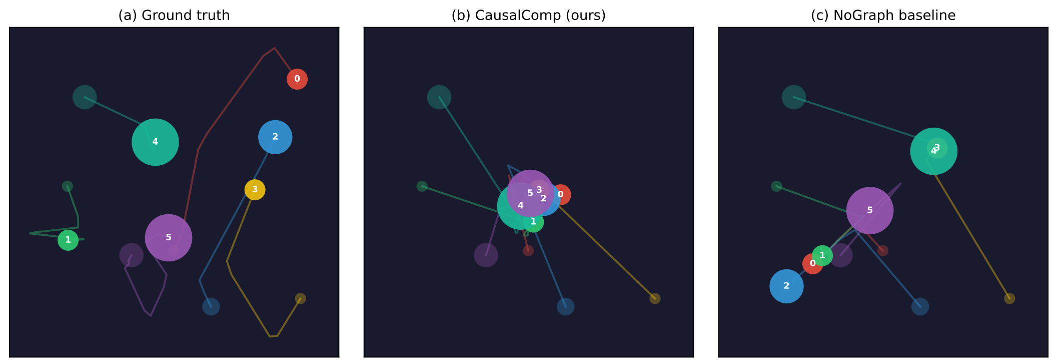

Figure 8: Trajectory prediction comparison. (a) Ground-truth trajectories of 6 objects over 16 frames. (b) CausalComp predictions closely track ground truth, including after collisions. (c) NoGraph baseline shows drift due to inability to model inter-object interactions. Filled circles show final positions; trails show full trajectories.

NeurIPS Paper Checklist

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   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract claims that typed interaction modules serve as a useful inductive bias for compositional transfer, supported by the SingleModule vs. CausalComp ablation across five environments (Tables 1–5), the capacity-matched ablation (Table 6), and the representation specialization analysis (Table 7). The abstract explicitly states we do not claim state-of-the-art prediction accuracy.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 5 discusses four limitations: (1) edge supervision uses GT collision events rather than unsupervised discovery, (2) all experiments use GT object states (including Physion), (3) M is a hyperparameter, and (4) no downstream task evaluation (planning / model-based RL).

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

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   Justification: Appendix A provides full architecture details (layer sizes, activation functions), training details (optimizer, learning rate, schedule, number of epochs), and environment specifications. All synthetic environments are procedurally generated from described parameters.

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   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

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   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

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   Justification: Section 4.1 describes environments, compositional splits, and metrics. Section 4.6 reports the hyperparameter sensitivity analysis for M. Appendix A provides full training details. Appendix B describes the baseline tuning protocol.

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   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

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   Justification: All results in Tables 1–7 report mean $\pm$ standard deviation over 3 random seeds. The seeds control both data generation and model initialization. Error bars capture variability from random data splits and parameter initialization.

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   Justification: Experiments were conducted on a single NVIDIA A100 80GB GPU. Each model trains in approximately 30–90 minutes depending on environment complexity. The full evaluation (5 environments $\times$ 5 methods $\times$ 3 seeds) requires approximately 80 GPU-hours total.

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    Justification: The NRI Springs benchmark is cited (Kipf et al., 2018). The Physion benchmark is cited (Bear et al., 2021). Our implementation follows the standard protocols. All other environments and code are original.

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    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [Yes]

    Justification: We introduce two new synthetic environments (SimplePhysics, MultiPhysics) and compositional evaluation splits. These are fully documented in Section 4.1 and Appendix A, with code provided as supplementary material.

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